



Azure Document Intelligence

Complete Integration Manual for ALGT ERP

Product	ALGT ERP — Document Management System (DMS)
Module	AI Settings → Azure Document Intelligence Provider
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This manual describes how to configure, integrate, and operate Microsoft Azure Document Intelligence within the ALGT ERP system for Arabic-optimized OCR, UAE document extraction, and bilingual AI processing.

1. Introduction

Azure Document Intelligence (formerly Azure Form Recognizer) is Microsoft's cloud AI service for extracting structured data from documents. It uses pre-trained and custom machine-learning models to read printed text, handwriting, and structured forms in over 100 languages — with exceptional support for Arabic.

The ALGT ERP system operates across UAE-based logistics, transport, offshore/onshore operations, HR, and government compliance workflows. Most UAE official documents — Emirates ID cards, trade licenses, labor contracts, and government certificates — are bilingual Arabic/English or Arabic-only. Standard PDF text-extraction tools often fail to read Arabic correctly due to right-to-left (RTL) encoding, bitmapped Arabic fonts, and custom character mappings.

Azure Document Intelligence solves these problems with a native Arabic OCR engine that correctly handles RTL text direction, Arabic Unicode, Tashkeel diacritics, and bilingual document layouts.

1.1 Why Azure Document Intelligence for Arabic?

✓ Native Arabic OCR	Trained on millions of Arabic documents including UAE government IDs and trade licenses
✓ RTL Text Direction	Reads right-to-left Arabic text correctly without character reversal
✓ Bilingual Layout	Handles mixed Arabic/English documents with correct reading order per column
✓ UAE ID Pre-trained	prebuilt-idDocument model is specifically trained on Emirates ID card layouts
✓ Arabic Handwriting	Recognizes Arabic handwritten signatures and annotations
✓ Tashkeel Support	Preserves Arabic diacritics (harakat) that distinguish word meanings
✓ Lower AI Cost	Per-page pricing is much lower than GPT-4.1 vision for large document batches
✓ Faster Processing	Async polling model returns results in 2-10 seconds vs 15-30s for GPT-4.1 vision

2. Prerequisites

2.1 Azure Account

You need an active Microsoft Azure subscription with billing enabled.

- Azure Portal: **portal.azure.com**
- Recommended region: **UAE North** (Dubai) or East US 2 for lowest latency
- Pricing tier: **S0 (Standard)** — required for pre-built models and Arabic support

2.2 ALGT ERP Access

- ALGT ERP Admin access with role: **system_admin** or **group_admin**
- Access to: **Admin** → **Settings** → **AI Settings**
- Access to the server **.env.local** file (or deployment environment variables)

2.3 Permissions Required in ERP

settings.ai.view	View AI provider configurations in Admin
settings.ai.manage	Add and edit AI provider configurations
dms.documents.upload	Upload documents through DMS
dms.admin	Run OCR and AI operations (or any role with dms.documents.ai.run)

3. Creating the Azure Resource

3.1 Step-by-Step Azure Setup

Step 1: Sign in	Go to portal.azure.com and sign in with your Azure account.
Step 2: Create resource	Click "+ Create a resource" → search for "Document Intelligence" → click Create.
Step 3: Configure	Fill in: Subscription, Resource Group (create new if needed), Region: UAE North, Name: algt-erp-doc-intelligence, Pricing tier: S0
Step 4: Review + Create	Click "Review + Create" → wait for validation → click "Create". Deployment takes 1-2 minutes.
Step 5: Get credentials	Go to the new resource → Keys and Endpoint. Copy: Endpoint URL and Key 1.

3.2 Recommended Azure Configuration

Resource Name	algt-erp-doc-intelligence (or any descriptive name)
Region	UAE North (Dubai) — lowest latency for UAE documents
Pricing Tier	S0 Standard — required for pre-built models
API Version	2024-11-30 (latest stable as of June 2026)
Endpoint Format	https://[resource-name].cognitiveservices.azure.com

■ Never share your Azure Key 1 or Key 2. Store them ONLY as environment variables. The ALGT ERP system stores only the variable name (secret_ref) in the database — never the actual key value.

4. Available Models

Azure Document Intelligence provides several pre-built models. Choose the right model based on your document type for best Arabic extraction accuracy.

Model ID	Best For	Arabic	Notes
prebuilt-read	Any document, Arabic PDFs, scanned letters, mixed bilingual docs	■■■■■	Best general Arabic OCR. Reads RTL text, handwriting, tables. START HERE.
prebuilt-idDocument	Emirates ID, Passport, Driving License, Residence Permit	■■■■■	Pre-trained on UAE ID card layouts. Extracts name, ID#, nationality, DOB, expiry automatically.
prebuilt-invoice	Tax Invoices, Arabic invoices, supplier bills	■■■■	Handles Arabic invoice RTL column structure. Extracts vendor, amount, VAT, line items.
prebuilt-document	Contracts, certificates, agreements, reports	■■■■	Extracts headings, paragraphs, tables, key-value pairs from Arabic structured documents.
prebuilt-layout	Complex Arabic tables, forms with mixed layout	■■■■	Best for Arabic forms with tables. Returns full page layout with bounding boxes.

RECOMMENDATION

Start with **prebuilt-read** as your default model. It handles all document types correctly for Arabic text. Create a second provider config with **prebuilt-idDocument** specifically for Emirates ID and passport processing.

5. Environment Variable Setup

Before configuring the provider in the ERP Admin, add your Azure key to the server environment. This keeps the API key secure — it never touches the database.

5.1 Local Development (.env.local)

Open the file `c:\dev\agt-erp\.env.local` and add:

```
AZURE_DOCUMENT_INTELLIGENCE_KEY=abc123...your_key_here...xyz
```

```
# Optional: if using a second config for ID documents
```

```
AZURE_DI_ID_DOCUMENT_KEY=abc123...your_key_here...xyz
```

5.2 Production / Staging Deployment

Add the environment variable to your deployment environment:

- Vercel: Project Settings → Environment Variables → Add Variable
- Docker: Add to docker-compose.yml environment section
- Server: Add to systemd service file or server .env file

5.3 Environment Variable Naming Convention

Variable Name	AZURE_DOCUMENT_INTELLIGENCE_KEY
Secret Ref in ERP	AZURE_DOCUMENT_INTELLIGENCE_KEY (must match exactly)
Value	Your Azure Key 1 from Keys and Endpoint page
Never store in DB	The ERP database stores only the variable NAME, never the value

■ After adding the environment variable, restart your Next.js development server (`npm run dev`) for it to take effect. In production, redeploy or restart the service.

6. Configuring in ALGT ERP Admin

Once your Azure resource is created and the environment variable is set, add the provider through the ERP Admin panel.

6.1 Navigate to AI Settings

- Open your browser and go to: **/admin/settings/ai**
- Sign in as **system_admin** or **group_admin**
- Click the **"Add Provider"** button in the AI Providers section

6.2 Provider Configuration Fields

Fill in the "Add AI Provider" form with the following values:

Config Code	ARABIC_OCR_AZURE	Unique identifier used in code
Provider Type	azure_document_intelligence	Must match exactly
Provider Name	Azure Document Intelligence (Arabic OCR)	Display name
API Endpoint	https://[resource].cognitiveservices.azure.com	From Azure portal
Model ID	prebuilt-read	Start with this; see Section 4 for options
API Version	2024-11-30	Latest stable version
Purpose	ocr	Select OCR from the dropdown
Secret Ref	AZURE_DOCUMENT_INTELLIGENCE_KEY	Must match your .env variable name
Is Active	ON	Toggle enabled
Is Enabled	ON	Toggle enabled
Requires Review	OFF	OCR output is for internal use

6.3 Setting the Secret Key

- After saving the provider, click the **"Set Secret"** button on the provider row.
- Paste your **Azure Key 1** in the dialog.
- Click Save — the key is encrypted/masked. The DB stores only a preview (first 4 chars + ***).
- The actual key is read from the environment variable at runtime.

6.4 Testing the Connection

- Click "**Test Connection**" on the provider row.
- The system sends a minimal test request to Azure and confirms the endpoint is reachable.
- A green checkmark appears if the test passes.

■ If Test Connection fails: verify your Azure endpoint URL is correct (ends in .cognitiveservices.azure.com), check that Key 1 is correct and not expired, and confirm the environment variable name matches Secret Ref exactly.

7. How Azure Document Intelligence Works in the ERP

7.1 Processing Architecture

Azure Document Intelligence is used as an alternative to GPT-4.1 Vision for document OCR. The system automatically detects whether Azure DI is configured and uses it for text extraction. GPT-4.1 Vision is then used for classification and metadata extraction using the extracted text.

1	User uploads document	DMS Upload Inbox → Click Upload & AI Fill
2	System checks Azure DI	getAzureDocumentIntelligenceProvider() checks for ARABIC_OCR_AZURE config
3	Azure DI OCR	Document sent to Azure → analyzeWithAzureOcr() → poll result → extract Arabic text
4	GPT-4.1 Classification	Azure OCR text passed to GPT-4.1 → classify document type + extract metadata
5	Draft created	AI-filled intake review form created with Arabic text correctly extracted
6	Human review	User reviews AI suggestions → Approve & Save → Document becomes active

7.2 Fallback Behavior

If Azure Document Intelligence is not configured or returns an error, the system automatically falls back to the existing GPT-4.1 Vision pipeline. This ensures no document upload is broken even if Azure DI is unavailable.

Azure DI configured + working	Uses Azure DI for OCR → GPT-4.1 for classification
Azure DI not configured	Falls back to GPT-4.1 Vision (existing behavior — unchanged)
Azure DI returns empty text	Falls back to GPT-4.1 Vision automatically
Azure DI request times out	Falls back to GPT-4.1 Vision, logs safe error code

8. Technical Details — Async Polling

Azure Document Intelligence uses an asynchronous processing model. When you submit a document, Azure returns a 202 Accepted response with an Operation-Location header. The system then polls this URL until the result is ready.

Submit request	POST to /documentModels/{model}:analyze → returns 202 + Operation-Location
Poll interval	Every 2 seconds
Maximum polls	25 attempts (50 seconds maximum wait)
Timeout	60 seconds total request timeout
Status values	notStarted → running → succeeded / failed
Text extraction	From paragraphs[] array (preserves RTL order) → fallback to pages[].lines[]

8.1 Supported Document Formats

- **PDF** — all types: digital, scanned, image-based, Arabic-encoded
- **JPEG / JPG** — photos of documents, ID cards, certificates
- **PNG** — screenshots, scanned images
- **TIFF / TIF** — high-resolution scans
- **BMP** — bitmap images
- **HEIF** — iPhone photos (common for UAE ID card photos)

■ **Maximum file size: 500 MB. Maximum pages: 2000 pages per document. For best Arabic OCR accuracy, use 300 DPI minimum resolution.**

9. Optional: Second Provider for Emirates ID

For maximum accuracy on Emirates ID cards and passports, create a second provider configuration using the `prebuilt-idDocument` model. This model is specifically pre-trained on UAE identity document layouts.

9.1 Emirates ID Provider Configuration

Config Code	ARABIC_OCR_AZURE_ID
Provider Type	azure_document_intelligence
Provider Name	Azure Document Intelligence (ID Documents)
API Endpoint	https://[resource].cognitiveservices.azure.com
Model ID	prebuilt-idDocument
API Version	2024-11-30
Purpose	ocr
Secret Ref	AZURE_DOCUMENT_INTELLIGENCE_KEY
Is Active	ON
Is Enabled	ON

9.2 What `prebuilt-idDocument` Extracts from Emirates ID

FirstName	First name — English and Arabic
LastName	Last name — English and Arabic
DocumentNumber	Emirates ID number (format: 784-XXXX-XXXXXXX-X)
DateOfBirth	Date of birth — Gregorian
DateOfExpiration	Card expiry date
Nationality	Nationality — English code (UAE, IND, PAK, etc.)
Sex	Gender (M/F)
CountryRegion	Issuing country (ARE for UAE)

10. Arabic OCR Quality Guide

10.1 Factors That Affect Arabic OCR Accuracy

Factor	Impact on Accuracy	Recommended Action
Image Resolution	Very High	Use minimum 300 DPI. 600 DPI for small text.
File Format	Medium	PDF > PNG > JPEG (avoid heavy JPEG compression)
Document Angle	High	Scan documents flat, <5 degree tilt
Lighting	High	Even lighting, no shadows over Arabic text
Paper Quality	Medium	Clean paper, no coffee stains or folds over text
Font Type	High	Printed fonts: excellent. Handwriting: good. Decorative: fair.
Color vs B&W;	Low	Both work well. Color adds context for stamps.

10.2 What Azure DI Handles Correctly for Arabic

- ✓ Arabic Unicode text — all Arabic character sets (U+0600-U+06FF)
- ✓ Right-to-left text direction — correct reading order preserved
- ✓ Tashkeel diacritics — ██████████ ██████████ ██████████ preserved when visible
- ✓ Mixed Arabic/English in same line — bilingual documents handled
- ✓ Arabic tables — RTL table structure preserved in extraction
- ✓ Eastern Arabic numerals — ██████████ correctly recognized
- ✓ Arabic dates — ██/██/████ recognized (note: Hijri vs Gregorian requires post-processing)

10.3 Known Limitations

- Very small Arabic text (<6pt) — may have reduced accuracy
- Highly decorative Arabic calligraphy — may have reduced accuracy
- Torn or water-damaged documents — standard limitation for all OCR
- Hijri calendar dates — extracted as text; Gregorian conversion requires post-processing (handled by GPT-4.1 prompt)

11. UAE Document Types — Recommended Models

	Emirates ID	prebuilt-idDocument	Extracts ID#, name (Ar+En), DOB, expiry, nationality. Pre-trained on UAE ID layout.
	Passport	prebuilt-idDocument	Extracts passport#, name, nationality, MRZ lines, DOB, expiry, issue date.
	Trade License	prebuilt-read	Reads license#, company name (Ar+En), activities, issuing authority, dates.
	Visa / Residency	prebuilt-idDocument	Extracts visa#, entry date, expiry, sponsor name, entry type.
	Labor Contract	prebuilt-document	Extracts parties, dates, job title, salary, terms. Key-value pair extraction.
	Tax Invoice	prebuilt-invoice	Extracts vendor, buyer, TRN, invoice#, date, amounts, VAT, line items.
	Tenancy Contract	prebuilt-document	Extracts landlord, tenant, property, rent, EJARI number, dates.
	Medical Certificate	prebuilt-read	General text extraction. Fitness result, dates, issuing authority.
	Insurance Policy	prebuilt-document	Policy number, coverage dates, amounts, insured party, conditions.
	Power of Attorney	prebuilt-document	Principal, agent, scope, dates, notary information.

12. Azure Pricing Reference (June 2026)

Azure Document Intelligence pricing is per page. Costs are significantly lower than GPT-4.1 Vision for high-volume document processing.

Model	Price per Page	Notes
prebuilt-read	\$0.001	0–500,000 pages/month
prebuilt-idDocument	\$0.001	0–500,000 pages/month
prebuilt-invoice	\$0.010	Higher due to complex extraction
prebuilt-document	\$0.001	0–500,000 pages/month
prebuilt-layout	\$0.001	0–500,000 pages/month
GPT-4.1 Vision (compare)	\$0.05–\$0.15	Per page estimate — 50–150x more expensive

■ Pricing based on Azure pricing page (June 2026) for UAE North region. The first 500 pages per month may be included in the free tier. Check portal.azure.com/pricing for current rates.

13. Troubleshooting

Issue: Test Connection fails

Causes:

API key not set in .env.local
Endpoint URL is incorrect
Azure resource is in a different region

Solution:

Check
AZURE_DOCUMENT_INTELLIGENCE_KEY in
.env.local. Verify endpoint exactly matches Azure
portal (no trailing slash). Restart Next.js server
after env changes.

Issue: Arabic text returns as question marks or boxes

Causes:

Using wrong OCR provider (pdf-parse instead
of Azure DI)
Document has non-Unicode Arabic encoding

Solution:

Ensure ARABIC_OCR_AZURE config is Active +
Enabled. Re-run OCR on the document from the
document record → OCR tab → Re-run OCR (AI).

Issue: Empty text returned from Azure DI

Causes:

Document is image-only with no content
File format not supported
Low image quality

Solution:

System automatically falls back to GPT-4.1
Vision. Check document quality (300 DPI
minimum). Verify file is not corrupted.

Issue: Operation timeout after 50 seconds

Causes:

Azure region is far from your server
Very large document (100+ pages)
Azure service degradation

Solution:

Consider switching to UAE North region for lower
latency. For large documents, split into batches.
Check Azure status at status.azure.com.

Issue: model_id error in Azure response

Causes:

Typo in Model ID field
Model ID not available in your Azure region

Solution:

Valid models: prebuilt-read, prebuilt-idDocument,
prebuilt-invoice, prebuilt-document,
prebuilt-layout. Check Azure region supports the
model.

Issue: Right-to-left text order wrong

Causes:

Using pdf-parse instead of Azure DI
Document has mixed RTL/LTR without proper
Unicode markers

Solution:

Azure DI handles RTL correctly. If still wrong after
Azure DI is active, the issue is in the downstream
processing. Report to ERP engineering team.

14. UAT Acceptance Checklist

Use this checklist to verify Azure Document Intelligence is correctly integrated in your ALGT ERP system.

✓	Azure resource created in UAE North region	S0 Standard tier
✓	AZURE_DOCUMENT_INTELLIGENCE_KEY set in .env.local	Env var name matches Secret Ref
✓	ARABIC_OCR_AZURE provider added in Admin → AI Settings	Config Code: ARABIC_OCR_AZURE
✓	Provider type is azure_document_intelligence	Not azure_openai or openai
✓	Model ID set to prebuilt-read	Default model for general Arabic
✓	Provider is Active and Enabled	Both toggles ON
✓	Test Connection passes with green checkmark	Azure key and endpoint valid
✗	Upload Arabic Trade License → OCR returns Arabic text	Run OCR (AI) on document
✗	Arabic company name visible in extracted text field	Full Arabic Unicode preserved
✗	Upload Emirates ID → Arabic name extracted	name_ar field populated
✗	Upload Arabic-only document → English summary generated	AI summary explains content
✗	Smart Links suggest correct party for Arabic company name	Arabic name matching works
✗	Fallback works: disable provider → GPT-4.1 runs instead	No errors when Azure DI off

■ Mark items as complete by verifying each in your test environment before enabling Azure Document Intelligence in production.

15. Quick Reference Card

Config Code	ARABIC_OCR_AZURE
Provider Type	azure_document_intelligence
Default Model	prebuilt-read
ID Model	prebuilt-idDocument (Emirates ID, Passport)
Invoice Model	prebuilt-invoice
API Version	2024-11-30
Env Variable	AZURE_DOCUMENT_INTELLIGENCE_KEY
Admin URL	/admin/settings/ai
Max File Size	500 MB
Max Pages	2000 pages
Timeout	60 seconds (25 polls × 2s interval)
Fallback	GPT-4.1 Vision (automatic if Azure DI unavailable)
Pricing	\$0.001 per page (prebuilt-read / prebuilt-idDocument)
Azure Portal	portal.azure.com → Document Intelligence
Azure Status	status.azure.com